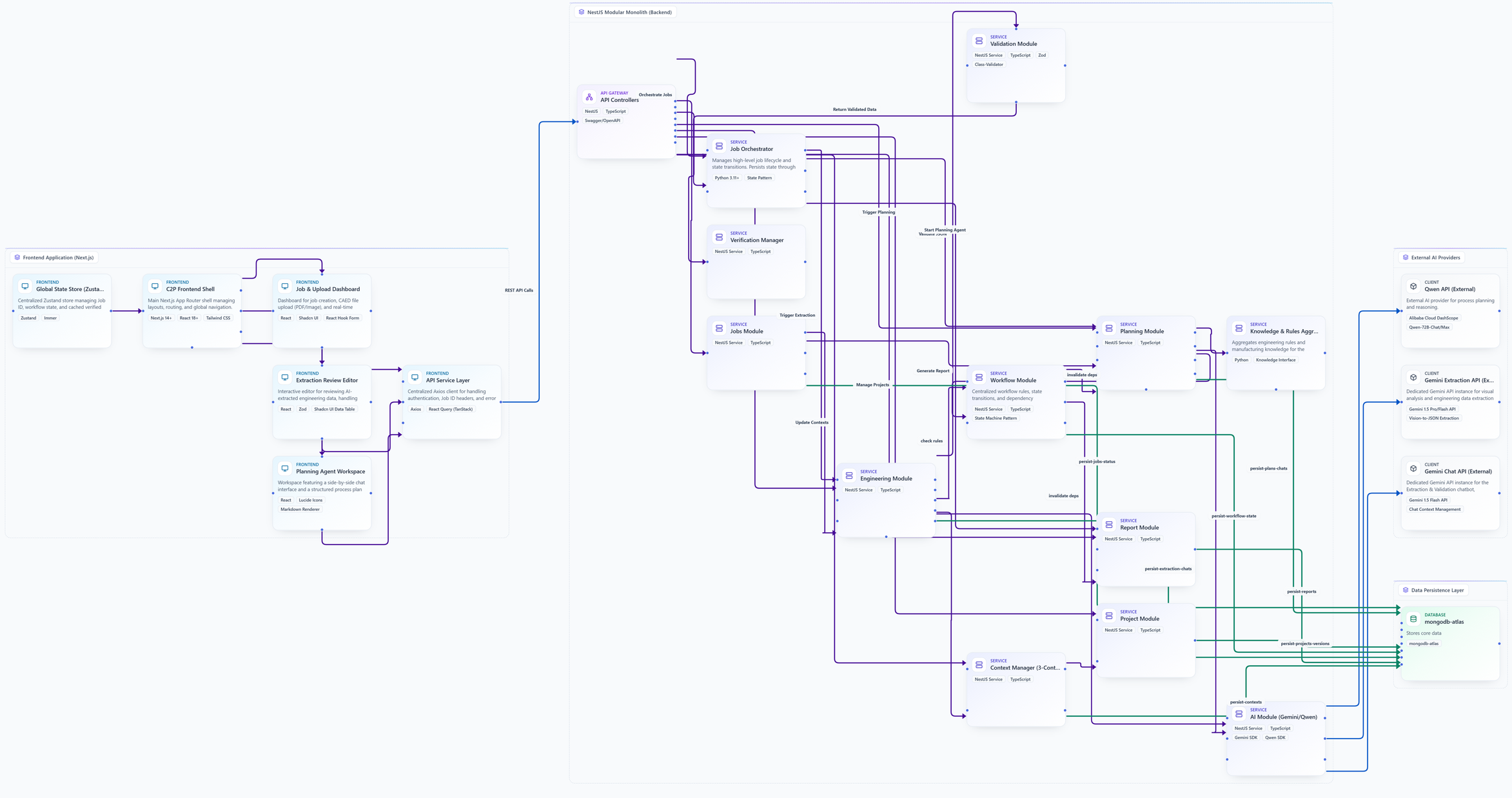
Frontend Application (Next.js)
FRONTEND
Global State Store (Zusta...
Centralized Zustand store managing Job ID, workflow state, and cached verified
Zustand
Immer
FRONTEND
C2P Frontend Shell
Main Next.js App Router shell managing layouts, routing, and global navigation.
Next.js 14+
React 18+
Tailwind CSS
FRONTEND
Job & Upload Dashboard
Dashboard for job creation, CAED file upload (PDF/Image), and real-time
React
Shadcn UI
React Hook Form
FRONTEND
Extraction Review Editor
Interactive editor for reviewing AI-extracted engineering data, handling
React
Zod
Shadcn UI Data Table
FRONTEND
API Service Layer
Centralized Axios client for handling authentication, Job ID headers, and error
Axios
React Query (TanStack)
FRONTEND
Planning Agent Workspace
Workspace featuring a side-by-side chat interface and a structured process plan
React
Lucide Icons
Markdown Renderer
REST API Calls
NestJS Modular Monolith (Backend)
API GATEWAY
API Controllers
Orchestrate Jobs
NestJS
TypeScript
Swagger/OpenAPI
SERVICE
Validation Module
NestJS Service
TypeScript
Zod
Class-Validator
Return Validated Data
SERVICE
Job Orchestrator
Manages high-level job lifecycle and state transitions. Persists state through
Python 3.11+
State Pattern
Trigger Planning
Start Planning Agent
SERVICE
Verification Manager
NestJS Service
TypeScript
Trigger Extraction
SERVICE
Jobs Module
NestJS Service
TypeScript
Manage Projects
Update Contexts
Generate Report
SERVICE
Planning Module
NestJS Service
TypeScript
invalidate deps
SERVICE
Knowledge & Rules Aggr...
Aggregates engineering rules and manufacturing knowledge for the
Python
Knowledge Interface
SERVICE
Workflow Module
Centralized workflow rules, state transitions, and dependency
NestJS Service
TypeScript
State Machine Pattern
check rules
persist-jobs-status
persist-plans-chats
invalidate deps
SERVICE
Engineering Module
NestJS Service
TypeScript
persist-workflow-state
SERVICE
Report Module
NestJS Service
TypeScript
persist-extraction-chats
persist-reports
SERVICE
Project Module
NestJS Service
TypeScript
persist-projects-versions
SERVICE
Context Manager (3-Cont...
NestJS Service
TypeScript
persist-contexts
SERVICE
AI Module (Gemini/Qwen)
NestJS Service
TypeScript
Gemini SDK
Qwen SDK
External AI Providers
CLIENT
Qwen API (External)
External AI provider for process planning and reasoning.
Alibaba Cloud DashScope
Qwen-72B-Chat/Max
CLIENT
Gemini Extraction API (Ex...
Dedicated Gemini API instance for visual analysis and engineering data extraction
Gemini 1.5 Pro/Flash API
Vision-to-JSON Extraction
CLIENT
Gemini Chat API (External)
Dedicated Gemini API instance for the Extraction & Validation chatbot,
Gemini 1.5 Flash API
Chat Context Management
Data Persistence Layer
DATABASE
mongodb-atlas
Stores core data
mongodb-atlas